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> #` Testing the MAPLE random number generator
> #According to the theory of uniform distribution
> #` Expected value  $\mu_X = \frac{(b + a)}{2}$ 
> #` Dispersion  $D = \frac{(b - a)^2}{12}$ 
> restart;
> a := 0; b := 1;
                                a := 0
                                b := 1
(1)

>  $\mu_{X\_T} := \frac{(b + a)}{2}$ ; #Theoretical average value
                                 $\mu_{X\_T} := 0.5000000000$ 
(2)

>  $D\_T := \frac{(b - a)^2}{12}$ ; #Theoretical value of dispersion
                                D_T := 0.08333333333
(3)

> Trn := 100;
                                Trn := 100
(4)

> #Trn = 1; 10; 100; 1000; 10000;
> rrm := evalf(  $\frac{\text{rand}(Trn \cdot a .. Trn \cdot b)}{Trn}$  );
rrm := 0.01000000000 proc( )
proc( ) option builtin = RandNumberInterface; end proc(6, 101, 7)
end proc
> rrm( );
                                0.9200000000
(6)

> Nexp := 100;
                                Nexp := 100
(7)

> # Nexp = 100; 1000; 10000; 100000; 1000000;
>  $\mu_{X\_Ex} := 0$ ;
                                 $\mu_{X\_Ex} := 0$ 
(8)

>
> for i from 1 to Nexp do
    rn[i] := rrm( ) :
     $\mu_{X\_Ex} := \mu_{X\_Ex} + \text{evalf}\left(\frac{rn[i]}{Nexp}\right)$  end do:
>  $\mu_{X\_Ex}$ ; #Calculated average value
                                0.5158000000
(9)

> D_Ex := 0;
(10)

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(10)

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> for  $i$  from 1 to  $N_{exp}$  do
   $D_{Ex} := D_{Ex} + \frac{(rn[i] - \mu_{Ex})^2}{N_{exp}}$  end do:
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> D_Ex;#Calculated dispersion value
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0.08924236000
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>  $\mu_{X_T} - \mu_{X_{Ex}}$ ; #Deviation from theoretical value
-0.0158000000
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> D_T - D_Ex;#Deviation from theoretical value
-0.00590902667

